



Association of Zoos & Aquariums

## 2023 AZA CONSERVATION GRANTS FUND

### Application Instructions

#### DEADLINE

This 2023 Conservation Grants Fund (CGF) electronic application must be used for all proposals. Completed applications must be submitted via email no later than **5:00 p.m. EDT** on **15 March 2023**. An auto-response email with a time-stamp will confirm receipt of the application.

*Incomplete applications, or applications received after 5:00 p.m. EDT on 15 March 2023, **regardless of cause**, will NOT be eligible for consideration. NO DEADLINE EXCEPTIONS WILL BE MADE.*

#### MEMBERSHIP

The Principal Investigator (PI) and all Co-Investigators (CIs) *must be individual AZA members* at the Professional Associate, Professional Affiliate, Professional Fellow, or Student level. Institutional membership does not count as individual membership and individual members at the Friend level are not eligible. Other participants actively involved in the completion of the project should be listed as collaborators and are not required to be AZA members. To apply for membership, visit <https://www.aza.org/individual-membership>.

#### FUNDING TIMELINE

All CGF projects must be completed within 12 months. CGF grants are intended to serve as seed money to attract additional funding and proposals should request funds only for this one-year period. Identical components of a project will not be funded for more than two grant cycles; however, funding requests for different components of the same project may be applied for in subsequent grant cycles.

CGF proposals undergo a two-tiered review process, outlined at <https://www.aza.org/cgf-selection-process>. Applicants will be notified of funding decisions in early September 2023 and funds will be available 2 October 2023. CGF funds can only be used for proposal expenses incurred on or after the award date. Expenses incurred prior to the award date will not be covered by CGF funds.

#### APPLICATION GUIDELINES

The application and all supplemental forms are posted online at: <http://www.aza.org/cgf-information-for-applicants/>. These documents contain form fields that must be completed and submitted via email to: [CGFapp@aza.org](mailto:CGFapp@aza.org). *All application materials must be in electronic format. No hard copies of any application materials will be accepted.*

It is strongly recommended that you save a copy of your completed application and supplemental forms as a backup. Only completed applications, submitted by the deadline defined above, will be reviewed. Applicants will be informed electronically if their application is incomplete.

**APPLICATION INSTRUCTIONS AND CHECKLIST:** All completed application materials must be submitted **in one email** by the deadline identified above and include the following:

**Application:** The application is one document consisting of the three sections outlined below. All sections must be complete in order for the application to be considered complete. The application must be submitted as a Microsoft Word document titled **"Application"** and attached to your Application email.

*Applications submitted in Portable Document Format (PDF) will NOT be eligible for consideration. NO EXCEPTIONS WILL BE MADE.*

**Note that “spell checker” and “track changes” functions cannot be used in this form.** AZA strongly recommends that applicants compose the content for their application in a separate document to make use of these features, and then copy and paste the final text into the form fields. When composing in a separate document, be aware of character limitations. If the copied text exceeds the character limit when pasted into a form field, *all text exceeding the character limitation will be cut off and will not appear in the final proposal.* Character limits include spaces.

Also note that certain formatting such as italics, bold, and underlining, is unavailable due to the application format. AZA recommends the use of numbering and spacing to emphasize important points in the text.

- ☒ **Cover Sheet.** Fill in the text fields for the project title, amount requested, and PI and CI information. Choose *two* project categories from the given list and identify the SAFE species that is (are) the focus of the project. Include a project abstract of 1500 or fewer characters.
- ☒ **Project Proposal.** Text fields are provided for responses to questions 1-12. Fields have a limit on the number of characters that may be used to keep the application concise. Character limits, including spaces, are stated following each question. If a question is not applicable to your proposal, write “N/A” in the field but *do not leave any form fields blank*, as this may delay processing of your application.
- ☒ **Budget.** Required budgetary information includes:
  - ☒ Detailed Project Budget. Detail how the budget will be allocated to specific items and indicate which funding source will pay for each item. State the total project budget, the amount requested from CGF, and other financial support applied for or received. Each of these amounts should reflect the one-year duration of the project. Funding decisions are announced in early September and CGF funds are to be used for work from the announcement date forward. Expenses incurred prior to the announcement will not be covered by CGF funds.
  - ☒ Budget Justification. Provide an explanation of why specific major items are essential to the project and reasonable in cost.

**Required Attachments:** The following must be attached to your Application email:

- ☒ **Curricula Vitae (CV).** A CV is required for each PI and CI listed on the Cover Sheet and must follow the format stated in these guidelines. CVs must not exceed two letter-sized (8.5 by 11 inch) pages each and must be typed in a legible font size (12 point recommended); each side of the paper counts as one page. CVs may be attached for collaborators but are not required. Each CV must be submitted as a separate document entitled, **“CV Investigator’s Last Name.”** CVs *exceeding the maximum length will be cut off at the second page.*
- ☒ **Statement of Institutional Support Forms.** Electronically signed *Statements of Institutional Support* from the CEO(s)/Institutional Director(s) (not the COO, General Curator, Vice President, Dean, or Department Director) of the PI’s institution and **each** CI institution *and* collaborating institution must be attached to the Application email as separate documents titled **“SIS Institution’s Name.”** The *Statement of Institutional Support* form can be found with the rest of the Application materials on the AZA website identified above.

Applicants wishing to include support letters from entities *other than* PI and CI institutions, collaborating institutions may submit such items as appendices.

Letters of support from PI and CI institutions and collaborating institutions will not be accepted in lieu of the *Statement of Institutional Support* form.

- ☒ **Statement of SAFE Species Program Alignment Forms.** Electronically signed *Statements of SAFE Species Program Alignment* from the program leader of relevant SAFE species program(s) must be attached to the Application email as separate documents titled “**SOA SAFE Species Program Name.**” The *Statement of SAFE Species Program Alignment* form can be found with the rest of the Application materials on the AZA website identified above.

Letters of support/endorsement from AZA Programs (Taxon Advisory Groups, Species Survival Plans®, SAFE) are **not** accepted, although coordination with AZA SAFE Programs is expected and should be described in Question 1. To assure program review of relevant submissions, AZA SAFE or Animal Program Leaders may serve as expert reviewers in the application review process and may express their support for a proposal at that time.

- ☐ **Permits.** If permits are required to complete your proposed project, you must attach a copy of each to your Application email. Each permit attached to the Application email must be titled “**Permit #.**” *If you are not able to obtain a required permit before submitting this proposal, you must submit it by 15 June 2023 to ensure award eligibility.*
- ☐ **Proposal Resubmission Form.** If a proposal for this project was previously submitted for AZA grant funding (i.e., CGF or SAFE granting program) and rejected, the revised proposal *must* include a completed *Proposal Resubmission Form* as an attachment on the Application email. This attachment must be submitted as a Microsoft Word document titled “**Resubmission Form**” and attached to your Application email. *Resubmission forms submitted in Portable Document Format (PDF) will not be accepted.* The form can be found with the rest of the Application materials on the AZA website identified above.

**Note: Proposals that fail to provide all items identified in the above checklist in the correct format will not be reviewed.**

**Non-Required Attachments:** The following attachments to your Application email are optional:

- ☒ **Citations (recommended).** Citations/references for statements made in the Application should be included in a separate document titled “**Citations**” and attached to the Application email.
- ☒ **Appendices.** Appendices are allowed, but may only *clarify*, and not *supplement*, information in the narrative. For example, photos or diagrams would be acceptable, but not videos, publications, or SAFE species program plans. Appendices must be attached to the Application email titled “**Appendix #.**”

**Note:** Appendices that do not adhere to these guidelines will be removed from the proposal prior to review.

## **SUBMISSION INSTRUCTIONS**

By submitting your application, you agree, if funded, to abide by the Grant Terms and Conditions posted on AZA’s website at <http://www.aza.org/cgf-information-for-applicants/>.

Proposals must be submitted electronically using the 2023 Application and related forms via a **single email, with the proposal title as the subject line**, to [CGFapp@aza.org](mailto:CGFapp@aza.org) no later than 5:00 p.m. EDT on 15 March 2023. Hard copies of any application materials or applications received after this deadline will not be accepted regardless of cause.

AZA's email system accepts messages up to 8 MB in size. In cases when applications are too large to be accepted in one email, AZA will accept the application submission in two separate emails, provided that the same proposal title is identified in the subject line of each.

The 2023 Application email **must** include the following attachments

- Application (this completed form);
- CVs for each Principal and Co- Investigator(s);
- Statements of Institutional Support for each PI and CI institution and collaborating institution;
- Statement(s) of SAFE Species Program Support; and
- A Proposal Resubmission form, if applicable.

The Application email may also include citation and appendices attachments.

**Auto-response emails will be sent to applicants confirming receipt of the application package. If an auto-response is not received, please resubmit the application. If an applicant submits more than one proposal, the person should ensure that an auto-response was received for *each* submission.** If submission problems arise, contact the AZA office at 301-562-0777. If you submit your application after the 5:00 p.m. deadline, you may still receive an auto-response from the Application inbox; however, your proposal will not be reviewed.

Applications which fail to follow submission instructions or are time-stamped after the deadline will not be considered and applicants will be notified that their application is inadmissible.

### **APPLICATION ASSISTANCE**

The CGF website (<http://www.aza.org/cgf-information-for-applicants/>) contains a wealth of information to help applicants compose a competitive proposal.

Proposals received before 8 March will be reviewed by AZA staff to ensure that the application is completed correctly. Early submission allows time for AZA staff to inform applicants if problems are noted in time for resubmission. Please note that AZA staff does not review early proposals for content.

## 2023 AZA CONSERVATION GRANTS FUND Cover Sheet

**Project Number:**  
For AZA use only

**Title of Project:** Maximizing fecundity through energy optimization: Is nutrition limiting egg laying habits of zebra sharks?

**Amount Requested:** \$20,060 (round to the nearest whole dollar)

**Project Start Date:** 9/30/23

**End Date:** 9/29/24

(CGF funding is for one year only, beginning no earlier than the award date)

**Investigators:** Each Principal Investigator (PI) and Co-Investigators (CI) *must be an individual AZA member at the time of application* at the Professional Associate, Professional Affiliate, Professional Fellow, or Student level. Other participants should be listed as collaborators on this form.

A Curriculum Vitae (CV) is required for each PI and CI, and a Statement of Institutional Support (SIS) form is required for each PI and CI Organization.

**1. Principal Investigator (PI) Name and Title:** Dr. Lance Adams (Senior Veterinarian)

**CV Attached?** ☒ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions, including affirmation that the organization is or will be a program partner of the relevant SAFE species program(s) upon acceptance of an award (required; available at: <http://www.aza.org/cgf-information-for-applicants/>). ☒

**Individual AZA Member Number:** 36300 (required)

**Institution/Organization:** Aquarium of the Pacific

**Statement of Institutional Support Form Attached?** ☒ (required)

**Institutional Address:** 100 Aquarium Way, Long Beach, CA, 90802

**Phone:** 562-951-1716

**E-mail:** ladams@lbaop.org

**2. Co-Investigator (CI) Name and Title:** Dr. Jennifer Wyffels (Research Scientist)

**CV Attached?** ☒ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions, including affirmation that the organization is or will be a program partner of the relevant SAFE species program(s) upon acceptance of an award (available at: <http://www.aza.org/cgf-information-for-applicants/>). ☒ (required)

**Individual AZA Member Number:** 81019250 (required)

**Institution/Organization:** Ripley's Aquarium

**Statement of Institutional Support Form Attached?** ☒ (required)

**Institutional Address:** 7576 Kingspointe Parkway, Suite 188, Orlando, FL, 32819

**Phone:** 865-393-4709

**E-mail:** wyffels@ripleys.com

**3. Co-Investigator (CI) Name and Title:** Dr. Kady Lyons (Research Scientist)

**CV Attached?** ☒ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions, including affirmation that the organization is or will be a program partner of the relevant SAFE species program(s) upon acceptance of an award (available at: <http://www.aza.org/cgf-information-for-applicants/>). ☒ (required)

**Individual AZA Member Number:** 81140495 (required)

**Institution/Organization:** Georgia Aquarium

**Statement of Institutional Support Form Attached?** ☒ (required)

**Institutional Address:** 225 Baker St NW, Atlanta, GA 30313

**Phone:** 310-387-7465

**E-mail:** klyons@georgiaaquarium.org

Information about additional co-investigators may be included in a separate document titled “**Additional Co-Investigators**” and attached to your Application email.

**Collaborating Institutions or Organizations:** A collaborating institution is defined as any institution actively involved in the design or completion of the project. Funding institutions *not* actively involved in project design or completion should be listed in the “Grant/Matching Support” section of the Project Budget and need not be listed as collaborators.

List all collaborating institutions below, including the institutions of the PI and CIs listed on the Cover Sheet, as well as those associated with the personnel listed in Question 6, and attach a Statement of Institutional Support form for each collaborating institution to the Application email.

Collaborating Institution Names:

Aquarium of the Pacific      *Statement of Institutional Support Form Attached?* ☒ (required)

Ripley's Aquarium      *Statement of Institutional Support Form Attached?* ☒ (required)

Georgia Aquarium      *Statement of Institutional Support Form Attached?* ☒ (required)

Fort Wayne Children's Zoo      *Statement of Institutional Support Form Attached?* ☒ (required)

Shedd Aquarium      *Statement of Institutional Support Form Attached?* ☒ (required)

Note: If additional Collaborating Institutions or Organizations need to be included, add their information in a separate document titled “**Additional Collaborators**” and include it as an attachment to your Application email.

**Project Categories (choose two that best describe the project):**

☐ Animal / Population Management (*Ex situ*)

☐ Conservation Education / Engagement

☒ Animal Health

☐ Field Conservation (*In situ*)

☐ Animal Welfare

☒ Research

☐ Communications / Public Relations

**Identify the SAFE species program(s) this project will benefit:** Safe Sharks and Rays

**Project Abstract (limit 1,500 characters with spaces):**

With the financial support of previous AZA funding, the reproductive biology of 16 zebra shark females from 7 institutions was characterized to generate important information about egg production for use by the *Stegostoma tigrinum* Augmentation and Restoration (StAR) Project. Notably, the total number of eggs laid and the proportion of yolked to wind (i.e. yolkless) eggs varied widely between females and institutions. Considering that environmental cues (i.e. light and temperature) were relatively consistent across AZA institutions, other factors may be responsible for the variability in egg production. The proposed project seeks to explore the relationship between female diet and fecundity to determine if energy limitation may account for the observed variation in egg production. This project will build upon the foundation generated from previous research to first leverage banked egg samples for bomb calorimetry and historic nutritional information on individual diet items to retrospectively compare energetic inputs (diet) with energetic outputs (eggs) for females across the egg laying spectrum (i.e. low-yolk to high-yolk producers). Second, a nutritional switch pilot study will be conducted to quantify the connection strength between diet and egg quality. Both results have important implications not only for animal welfare and management but also conservation applications for the StAR initiative, whose success is inextricably linked to egg production.



## 2023 AZA CONSERVATION GRANTS FUND

### Project Proposal

1. Describe the overall project and its significance for conservation and/or management. What species, sites, habitats, populations, or questions are targeted or addressed and what critical needs will be met? Explain the project's relevance to the SAFE species' program team's priorities and how it supports implementation of the team's 3-year program plan and existing regional and/or global recovery plans. **LIMIT 5,000**

#### **CHARACTERS with spaces**

The zebra shark (*Stegostoma tigrinum*; herein "Steg(s)") is listed as endangered by the IUCN and are a AZA SAFE designated species. In 2019, the *Stegostoma tigrinum* Augmentation and Restoration (StAR) Project was conceived with the goal to repopulate Indonesian waters where Stegs have been extirpated by exporting surplus Steg eggs produced by females at AZA and partner institutions. On January 13th, 2023, StAR released the first 2 young-of-year Stegs to Indonesian waters that resulted from eggs hatched and reared in-country shipped from SeaLife Sydney, demonstrating proof of concept for the reintroduction project. Additional shipments been sent to the Indonesian hatcheries with the intent to scale up the number of sharks released and number of institutions contributing eggs. A Population Viability Assessment (PVA) forecasted that anywhere from 64 (pessimistic) to 100 (optimistic) fertile eggs per year need are needed to meet conservation goals (Traylor-Holzer 2021).

To assist with egg shipment planning, a reproductive study was conducted to characterize egg laying patterns in this species, supported by the AZA Conservation Grant Fund (awarded in Fall 2021). Egg laying records were collated for 16 females from 7 collaborating institutions for one year (Appendix A). That study documented a wide range in both the total number of eggs and the proportion of yolked to wind (i.e. yolk absent) eggs laid annually (Appendix B). For example, at Georgia Aquarium (GAI) in 2021, only 22% of eggs laid by 3 resident females were yolked. By contrast, 86% of eggs laid by a female at Golden Nugget Las Vegas (GNLV) were yolked. The lack of efficient yolked egg production poses a challenge for institutions that wish to participate in the StAR program, but also for the program itself to meet the quotas recommended by the PVA report.

Environmental cues can influence elasmobranch reproduction (Mull et al. 2008). In most US institutions, both light and temperature are kept relatively constant year-round including collaborating institutions in the reproductive study. For example, both GAI and GNLV aquaria are exposed to natural light via skylights and maintain water temperature at 76.1F (24.5C). Although age could be an important factor affecting female fecundity and egg quality, females at both institutions have been in human care for similar lengths of time (GAI = 14-16 yrs, GNLV = 13 yrs), suggesting that other factors play an important role in determining individual fecundity.

Unlike most oviparous shark species, the Steg egg is considerable both in its dimensions and mass, with yolks ranging from 22 to 110 grams (Appendix C). The production of such a large yolk mass suggests that the energetic investment to create each egg is substantial. In preparation for and during laying periods, females shift internal (liver) energy reserves to the ovary where yolk accumulates in follicles. Females likely have increased and/or different energy requirements during egg laying periods than during reproductively quiescent periods. However, most institutions feed a constant diet throughout the year, often determined as a percentage of shark weight. At Fort Wayne's Children Zoo, IN, the aquarists have noted a change in preferred diet items by their female during laying periods (V. Hunter, personal observation), suggesting that when possible, females may choose to modify their diet to accommodate energetic demands related to reproduction. However, the energetic cost for a female to produce an egg is unknown, making it difficult to determine if the energy available is sufficient to meet the high caloric demands of egg laying.

In other species, nutritional deficiencies can negatively impact reproductive signals, including those that trigger ovulation (Wade et al. 1996), effectively interfering with the normal reproductive process. In an oviparous species, ovulation and egg case production must be temporally coordinated. If signals to the ovary and shell gland became disconnected or out of synchronization, a situation where egg cases are produced by the shell gland but follicles are not ovulated at the same rate and would result in wind eggs. However, studies examining the connection between diet and reproduction in elasmobranchs are limited, and ex situ studies have focused on maintaining a healthy body weight year-round rather than variable energetic demands associated with reproductive processes. Therefore, the aim of this study is to explicitly investigate the relationship between female diet and egg production. Information from this study will provide baseline information to edify and guide diet-related animal welfare best practices for elasmobranchs, and has significant application for the StAR program that is dependent on the efficient production of yolked eggs to meet conservation goals.

2. List the goal(s) and associated objective(s) of the proposed project and the actions that will be taken to achieve them. Goals should describe the desired status in the long-term of the question or problem that the project will address. Objectives should state the immediate results that are necessary to meet the long-term goal(s). Actions should be the specific activities the project will undertake to complete each objective. If the proposal is a request to fund part of a larger project, this section should focus on the part for which CGF funding is being requested during the one-year duration, while also providing some context regarding the larger effort. **LIMIT 4,500 CHARACTERS with spaces**

The mission of the StAR project is to re-establish a healthy, genetically diverse and self-sustaining population of Stegs in the Raja Ampat Archipelago, Indonesia. Viable eggs will be transported to Raja Ampat to complete incubation in one of two new facilities dedicated to this conservation cause. Thus, Stegs at aquaria worldwide can serve as animal ambassadors for their species as well as directly contribute to re-introduction efforts. The success of this pioneering project is first and foremost dependent on the successful production of yolked eggs in quantities necessary to meet the targets outlined by the PVA (Traylor-Holzer 2021). Thus, there is pressing need to identify factors that may be limiting yolked egg production.

Because Steg eggs have large yolks that presumably are energy dense, we hypothesize that female diet could be a limiting factor. Diet has an obvious biological role in reproduction and represents a factor that could substantially vary across institutions. Previous work showed that periods of oviposition vary by female from short (2 months) to long (4-5 months) laying seasons. Since diets are kept fairly constant, females may be energetically taxed by the end of a reproductive season and could lay eggs of lower quality (i.e. less energy) or stop laying due to an energy deficit. Therefore, in addition to supporting conservation efforts, this work has animal welfare implications underscoring the need to understand if the energetic costs to produce each egg are appropriately being matched with diet.

Outlined below are the goals and objectives necessary to determine the relationship between Steg diet and fecundity as well as egg quality.

Goal 1: Determine the energetic cost of egg production over 1 year

Objective 1: Quantify the energy content of an egg

Action 1a: Measure the caloric values of each egg component using bomb calorimetry from historic samples

Action 1b: Use information from 1a to calculate a female-specific energetic cost for eggs laid during one year

Objective 2: Determine if female investment for each egg varies during the egg laying season

Action 2a: Analyze trends in egg energetics over time for one year

Objective 3: Assess variability in female egg laying energetics within and across institutions

Action 3a: Compare the total egg energy investment during one year for females within and across institutions

Goal 2: Characterize corresponding Steg diets over 1 year

Objective 1: Describe the Steg diet that corresponds with egg laying (Objective 1)

Action 1a: Survey participating institutions for quantity and identity of Steg diet items consumed

Objective 2: Examine if female food consumption patterns change over the course of one year

Action 2a: Compare total food consumed and proportion of food items over time

Objective 3: Quantify diet compositions and consumption variability

Action 3a: Compare types and quantities of female diets within and across institutions

Goal 3: Correlate diet with egg production energetics for Stegs with opposing yolked-egg laying habits

Objective 1: Explore correlation between diet and egg laying patterns

Action 1a: Overlay food consumption parameters with egg laying behavior

Objective 2: Calculate an annual energy budget for each female

Action 2a: Compare energetic inputs (diet) with energetic outputs (egg laying behavior and egg quality) among females to understand if diet limits the production of yolked eggs for Stegs.

Goal 4: Quantify the degree of connectivity between nutrition and egg quality

Objective 1: Change diet of co-housed females

Action 1a: Feed co-housed females calorically equivalent and nutritionally distinct (fish or invertebrate dominated) diets for one quarter of the year. Revert to original diet for the remaining three quarters.

Objective 2: Quantify egg nutrients pre- and post-diet switch

Action 2a: Collect and analyze eggs for six months, beginning at the onset of diet change and for one quarter after reverting to the original diet

Objective 3: Determine if changes diet are reflected in yolk nutritional status

Action 2: Compare egg nutrients throughout the six-months of diet changes as well as to historical data (goal 1)

3. Explain in detail the methodology for this project and why it is appropriate. If this is a research project, clearly state the research question(s) and how it will be tested. For applied conservation projects, explain how project actions will be undertaken and why the proposed actions are appropriate for addressing the conservation issue identified. If this is a conservation education or professional training project, describe the methods as they relate to the target audience, number of people that will be reached, etc. **LIMIT 7,000 CHARACTERS with spaces**

#### Egg Energetics

##### Egg Sample Collection

From previous work supported by the AZA Conservation Grant Fund, we (JW, KL) monitored the laying activities of 16 females at 7 institutions. As part of that research, eggs (both wind and yolked) were dissected to obtain the mass of various components prior to banking samples for later analysis. Wind eggs and yolked eggs have the same components, save for the presence or absence of a yolk, although the proportions of each egg component vary. The energy needed to create an egg can be broken into three main components: shell, supporting internal material and yolk (Appendix D). Because the internal, non-yolk material is mostly water (Wyffels, Lyons unpublished data), sample composites were taken by pooling materials from wind eggs from the same female and frozen to ensure sufficient amounts of organic material were available for bomb calorimetry. Shell wet mass was measured and the shell subsequently dried to calculate percent organic material. A subset of dried egg shells were frozen. In non-wind eggs, yolk total mass was measured and the yolk homogenized prior to freezing 1-2mL subsamples.

##### Calorimetry

Bomb calorimetry is a technique long used to determine the total energy content stored in organic material. The total energy of an item can be accurately quantified based on the change in water bath temperature as a result of sample combustion. We will use bomb calorimetry to obtain caloric values (energy per gram of tissue) for each egg component (shell, internal material, yolk).

Energy content of each egg will be calculated as follows:

$$(\text{Mass of internal components} \times \text{calories/gram}) + (\text{mass of yolk} \times \text{calories/gram}) + (\text{mass of shell} \times \text{calories/gram}) = \text{total egg energy}$$

The energy content per gram of internal material and egg shell is assumed to be similar among females, although the mass of each component varies. Average caloric values for egg shell and internal material will be calculated as the mean of composites contributed by 5 females. By contrast, because yolk is expected to contribute the most to total egg energy, with greater potential to vary between females and across a laying season, yolks will be individually analyzed. Half of all banked yolk samples will be bombed from each female with samples spanning the female's laying cycle as determined from egg laying records (Appendix A).

##### Biological Variance

Egg total energy will be examined longitudinally by female to identify changes in energy investment throughout the laying season. Total energy spent in a laying season will be calculated as the sum of all egg energy output and compared both within and across institutions.

##### Diet Characterization

While Steg diet descriptions have been compiled from previous AZA surveys (Watson and Janse 2017), we aim to take a more thorough and individualized approach specific to the females in the study. Institutions will be asked to provide daily/weekly food logs for each female and vendors used during the year egg laying habitats were monitored as part of the reproductive research. Food logs will include the type and mass of each food item offered, amounts consumed, time and days animals were fed, whether animals have access to broadcast feeding and if diet was supplemented with vitamins.

Amount of food (relative to body size where data are available) and proportion of items offered/consumed will be compared across institutions on a weekly basis. For each female, diet will be compared during laying and non-laying periods as well as between females at the same institution, and across institutions.

Nutritional information from a decade of diet analysis at GAI suggests that individual food items have a relatively constant caloric content over time (Dr. Lisa Hoopes, personal observation). In addition, because most AZA institutions use the same vendors, analyzing individual food items from each institution is not necessary. Therefore, the energy content will be calculated for each diet item using historic records as a reliable proxy to determine total caloric input provided to females each week. The egg energy output (goal 1) will be subtracted from the estimated total dietary calories provided each week and compared for females during laying and non-laying periods as well as among institutions. If institutions

have sourced diet items from a vendor not represented in historic records, 3 samples of each food item will be collected for bomb calorimetry.

#### Energy balance between dietary input and egg laying output

Information collected as part of the previous AZA CGF proposal demonstrate that females span a spectrum of egg laying habits from low-yolked to high-yolked producers. This knowledge will be leveraged to more closely examine if dietary differences exist between females at these institutions and how diet may (or may not) be related with regards to yolked egg output.

Total calories provided by weekly diet (input) will be compared with total calories expended on egg laying (output) between low-yolk and high-yolk producing females at each institution. We predict that high-yolk producing females will have higher caloric diets than low-yolk producing females.

#### Diet shift study

While Steg females are hypothesized to expend a substantial amount of energy into creating eggs (goal 1), it is unknown how tightly coupled this reproductive investment is to the quality of their diet. While Stegs in situ likely primarily feed on invertebrates (Christine Dudgeon, personal communication), many animals in human care are fed teleost fishes that include species from localities that would never occur naturally in their diet. For example, herring and capelin are dietary staples in the industry, however Stegs would not have access to these Arctic species in situ. Furthermore, Arctic prey fishes tend to be high in omega-6 fatty acids that are correlated with inflammatory responses (Lisa Hoopes, personal communication). To determine if changing diet items reflects in increased yolk quality, we will conduct a pilot diet switch experiment at Aquarium of the Pacific as proof of concept. Two co-housed females will be fed calorically equivalent and distinct diets: one that is primarily fish-based and one that is primarily invertebrate-based. Females will be fed their new diet for one quarter and animal welfare will be monitored for behavioral and body condition changes. Eggs from females will be collected for six months beginning at the onset of diet change. The monitoring period includes 3 months in parallel with diet change and 3 months after returning to the original diet. Yolks for the diet study will be analyzed for total energy, crude protein, fat and ash content.

4. If AZA has provided previous funding for this or a related project through its grants programs, summarize relevant findings and identify how they relate to the current proposal. **LIMIT 1,000 CHARACTERS with spaces**

Findings from research previously supported by the AZA CGF directly lay the foundation for the work proposed here. In the previous proposal, we documented the laying behaviors of 16 females across 7 institutions over the course of one year. An unexpected finding of that research was the variability in both the number of eggs laid and proportion of wind to yolked eggs (Appendix B). The second goal of that work described baseline rates of parthenogenesis. We discovered that genetically-confirmed parthenotes were produced by every female examined in the study.

The current proposal is inextricably linked with the previously funded work. Only by conducting the previous project did we discover a clear need for further research to understand factors that may be responsible for the low fecundity observed at certain institutions compared to others. A significant "head start" will be afforded to the current proposal as most samples for analysis were banked during the reproduction research.

5. Describe the timeline in which components of the project will be completed within the one-year project duration. Use of bulleted/numbered lists is encouraged. **LIMIT 1,500 CHARACTERS with spaces**

As a majority of egg samples were collected from previously funded work, the proposed project will complete its outlined objectives within a one-year timeframe (Appendix E) as follows:

#### Goal 1: Egg energetics

Sep '23—Select yolk samples from each female distributed across her laying season.

Oct '23—Send historically-collected samples of egg components (yolk, internal material, and shells) for analysis

Jan '24—Receive final egg bomb calorimetry data

Feb '24—Calculate egg energetics

May '24—Analyze trends in egg energetics for each female over one year

#### Goal 2: Diet Characterization

Sep '23—Request diet records from each female for the specified 1-year laying period

Jan '24—obtain historic diet information

Feb '24—Request frozen items originating from non-traditional vendor sources

Mar '24—Send food items for bomb calorimetry

May '24—Receive final diet bomb calorimetry data

Jul '24—Calculate weekly diet energetics using consumption data and caloric content

Goal 3: Energetic comparisons

Jul-Aug '24—Compare energetic inputs (diet) with energetic outputs (egg laying)

Goal 4: Diet switch pilot study

Sep '23—Work with husbandry team to procure diet items

Oct '23—Begin diet switch experiment; collect and bank yolk samples for 3 months

Jan '24—Revert to original diet; collect and bank yolk samples for 3 months

Jan-Apr '24—Send yolk samples for nutritional analysis

May '24—Receive yolk nutritional data

Jun-Jul '24—Analyze trends in egg and diet energetics and nutrition for eac

6. Define the project team by identifying the personnel included for this project, their roles, qualifications, and the key skills they bring. Discuss the PI and each CI listed on the Cover Sheet and any collaborators (persons actively involved in the completion of the project). Each collaborator's institution should be listed under **Collaborating Institutions or Organizations** on the application Cover Sheet. **LIMIT 2,500 CHARACTERS with spaces**

Aquarium of the Pacific – Dr. Lance Adams (DVM): Dr. Adams completed a veterinary internship in aquatic animal medicine and has 21 years of clinical aquatic veterinarian experience at Aquarium of the Pacific. Fourteen years of experience investigating artificial insemination in elasmobranchs, including successful artificial insemination of White Spotted Bamboo, Brown Banded Bamboo, and Zebra Sharks. Specific skills and experience with ultrasonography for the examination and evaluation of reproduction in elasmobranchs and methods for artificial insemination in oviparous sharks. Dr. Adams serves on the StAR Veterinary Group.

Ripley's Aquariums – Jennifer Wyffels (PhD): Dr. Wyffels has more than 20 years of experience researching elasmobranch physiology, in particular reproduction and embryonic development. She has both initiated and led multi-institutional collaborative research projects that include aquariums, government agencies and academia. Her position as a Research Scientist at Ripley's Aquariums provides a platform to highlight aquarium-led research activities with institutional guests and AZA community. Dr. Wyffels investigated baseline rates of parthenogenesis as part of the reproductive research and banked yolk samples that will be analyzed as part of the proposed research. She serves on the StAR Husbandry Group and is committed to the overall research objectives and their completion.

Georgia Aquarium – Kady Lyons (PhD): Dr. Lyons has over 10 years of experience researching elasmobranchs, in particular aspects of reproductive biology. She has a strong track record of carrying projects from start to finish as well as working in multi-institutional collaborative groups. Her position as a research scientist at the Georgia Aquarium will help highlight the research aspects being led by AZA member institutions. In 2019, Dr. Lyons was responsible for organizing and coordinating the Steg workshop that has now evolved into the StAR project. Thus, she has been closely involved in the StAR project since its inception. Finally, Dr. Lyons is a member of both the StAR Research and StAR Husbandry Group and, therefore, is closely invested in the overall project and committed to ensuring its success.

7. Describe the project's monitoring and evaluation plans. Explain how the goals and objectives of the project will be met, which indicators will be used, and how those indicators will be measured. If this project has an educational or professional training component, the educational impact should be discussed (ideally including impacts on conservation-related knowledge, attitudes/affect, and behavior). Plans may be quantitative or qualitative, and should differentiate between monitoring (ensuring proposed actions are taken) and evaluation (whether the project had the desired impact on the issue or question addressed).

**LIMIT 3,000 CHARACTERS with spaces**

Project leads will ensure open and regular communication with collaborators occurs throughout the project. This communication is necessary to ensure that collaborators provide historic diet data in a timely fashion, and to notify project leads when females at any of the four institutions begin their laying period. The following metrics will be used to measure demonstrable progress towards project goals and objectives:

Monitoring:

- Lead PIs will check-in electronically with collaborating institutions monthly to ensure that historic diet data is being collated and subsequently shared.
- Lead PIs will check-in electronically with collaborating institutions monthly to ensure that food items are being frozen and subsequently shared.
- Lead PIs will work together to select banked samples for analysis and coordinate shipping to a commercial lab for analysis
- Lead PIs will request progress updates monthly from analysis laboratories

Evaluation:

Goals will be evaluated based on achieving the following metrics/goalposts/markers. Contingency plans are also included in the description:

Goal 1: Egg energetics

- Identification and shipment of historic egg samples to commercial lab within the first month of the study
- Receipt of bomb calorimetry data within 4 months of the study. Several service-based bomb calorimetry laboratories have been identified ensuring timely processing of project samples.
- Calculation of total egg energy from bomb calorimetry and dissection data to be completed within one month of receipt of bomb calorimetry data

Goal 2: Diet characterization

- Receipt of detailed historic diet information within 4 months of the study. If institutions are found to have kept scant or no diet information, detailed records will be for the 2023-2024 cycle of the proposed work.
- Submit food items for bomb calorimetry from all institutions by month 6 of the study
- Receipt of bomb calorimetry data by month 8 of the study. Several service-based bomb calorimetry laboratories have been identified ensuring timely processing of project samples.
- Calculation of total diet energy from bomb calorimetry and institutional data to be completed within one month of receipt of bomb calorimetry data

Goal 3: Compare energetic inputs (diet) with energetic outputs (egg laying)

- Calculation and comparison of total energy measured from eggs and diets during months 10-11 of the study

Goal 4: Diet switch study

- Identify vendors and sources for prey items appropriate for the diet study
- Conduct diet switch study and monitor females for changes in behavior and condition
- Collect and confidently assign all eggs laid prior, during and after diet switch experiment
- Submit and receive nutritional data back from the laboratory by month 6 of the project.

Overall study goal:

- Data analysis and report preparation during month 12 of the study

8. Explain and identify the means and timeline of how the key results and lessons of the project will be disseminated and shared. Key internal and external audiences should be identified. **LIMIT 1,500 CHARACTERS with spaces**

Key results and lessons relevant to the success of the ongoing project will be communicated through a variety of avenues. Currently, Dr. Wyffels and Dr. Lyons provide regular communication to the StAR Steering Committee on reproductive-related aspects of the project. Therefore, a framework of communication has already been established with leadership of the StAR project.

Husbandry staff at participating institutions represent a key stakeholder in this project. Results will be shared internally with project collaborators at regular intervals and leverage the open communication channels that have been established with institutions from the work previously funded by the 2021-2022 AZA Conservation Grant Fund. One-on-one discussions and/or lunch-and-learns will be held with husbandry staff at each institution to interpret results and discuss potential adaptations to female diets.

Findings of the study will be broadly disseminated through publication in peer-reviewed, open-access journals within 6 months of completing project objectives. Project findings will be presented at conferences relevant to the industry such as the International Aquarium Congress and the Regional Aquatic Workshop, but may also be shared at meetings attended by professional fisheries biologists (e.g. the American Fisheries Society), as findings have broad relevance for not only husbandry of animals in human care, but also understanding fundamental concepts in fish biology.

9. Have all the necessary permits been obtained for the project?

a) ☐ Yes ☒ No ☐ Not Applicable

If yes, list the permits here and attach one copy of each to your Application email. Attachments should be titled **"Permit #."**

pending - currently being applied for

- b) If the necessary permits have not yet been obtained, please explain why. **Pending permits must be obtained and submitted to AZA by 15 June 2023.**

We are in the process of applying for institutional permission from Aquarium of the Pacific's Research Review Board. We have verbal institutional support for the diet shift and are in the process of filling out the appropriate paper work. We will also be submitting IACUC forms through Georgia Aquarium in the coming weeks.

10. If research animals are to be purchased for the project, indicate and justify the destination of the animal(s) after completion of the research. **LIMIT 750 CHARACTERS with spaces**

NA

11. If this is a multi-year project, identify the funding strategy for subsequent years. **LIMIT 750 CHARACTERS with spaces**

NA

12. **Past AZA Grant Support.** If Principal or Co-Investigators listed on the Cover Sheet have received previous AZA grant support (i.e., CGF or SAFE granting program), complete the following (required).

a) Name of Investigator: Kady Lyons Year of Award: 2021 Amount of Award: \$20,285  
Project Title: Characterization of adult zebra shark reproductive cycles and baseline rates of parthenogenesis  
Were all required reports (annual/final) submitted?  
☒ Yes ☐ No ☐ Not yet due / Other (briefly explain)

b) Name of Investigator: Year of Award: Amount of Award:  
Project Title:  
Were all required reports (annual/final) submitted?

☐ Yes

☐ No

☐ Not yet due / Other (briefly explain)

Note: If additional awards need to be included, add their information to a separate document titled “**Past AZA-Funded Projects**” and include it as an attachment with your Application email.

## 2023 AZA CONSERVATION GRANTS FUND

### Project Budget

#### BUDGET CYCLE

All funding requests from CGF must be for a maximum period of one year only. Funding decisions are announced in early September and expenses incurred prior to the award date will not be covered by CGF funds. CGF funds are to be used within one year from the project start date and forward.

#### DETAILED BUDGET

A detailed budget sheet is provided below and information should be separated into CGF and non-CGF costs. List under each category the budget items and method of calculation for each (e.g., 2 persons x 10 days x \$25/day), as well as probable sources of support. The following should be listed as **separate** budget items:

- airfares
- lodging
- vehicle rental (automobiles, planes, boats)
- fuel
- per diem
- equipment over \$500
- supplies, services (lab analyses, editing)
- pharmaceuticals
- shipping/postage
- printing
- personnel costs

**CGF does not fund the following:** Tuition, university fees, or fringe benefits associated with graduate students or other project personnel; administrative costs (including institutional overhead and submission costs for the publication of journal articles); travel unrelated to completion of the project (conference presentation expenses, travel to meetings, etc.); planning meetings that define, rather than implement, conservation goals.

**CGF does not fund salaries for project personnel other than graduate students, technicians, and start-up positions established solely for the purpose of completion of the project.**

**Note:** Recipients of CGF grants are obligated to inform AZA (CGF@aza.org) of any pending or overlapping grants received. If such grants are received, AZA's Scientific Advisory Committee may recommend that the award amount be reduced or refunded.

**Note:** If additional budget items need to be included, add this information in a separate document titled "**Additional Budget Items**" and include it as an attachment with your Application email.

**Budget Items Form:**

| Budget Item                                                                     | Method of Calculation<br>(e.g., 2 persons x 10 days x \$25/day) | CGF- Requested Amount | Non-CGF Requested Amount<br>(if applicable) | Date Expected from Non-CGF Source (if applicable and not yet received) | Name of Non-CGF Source (e.g., Grant Fund, Matching) | Total Expense (CGF+Non-CGF) |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------|---------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------|-----------------------------|
| AOP historic yolk samples (bomb calorimetry)                                    | 104 yolk samples x \$44/sample                                  | \$4,576               | \$                                          |                                                                        |                                                     | \$4,576                     |
| CZA historic yolk samples (bomb calorimetry)                                    | 36 yolk samples x \$44/sample                                   | \$1,584               | \$                                          |                                                                        |                                                     | \$1,584                     |
| FWCZ historic yolk samples (bomb calorimetry)                                   | 15 yolk samples x \$44/sample                                   | \$660                 | \$                                          |                                                                        |                                                     | \$660                       |
| GAI historic yolk samples (bomb calorimetry)                                    | 25 yolk samples x \$44/sample                                   | \$                    | \$1,100                                     | FY 2024                                                                | Georgia Aquarium                                    | \$1,100                     |
| GNLV historic yolk samples (bomb calorimetry)                                   | 28 yolk samples x \$44/sample                                   | \$1,232               | \$                                          |                                                                        |                                                     | \$1,232                     |
| RIP historic yolk samples (bomb calorimetry)                                    | 7 yolk samples x \$44/sample                                    | \$                    | \$308                                       | FY 2024                                                                | Ripley's Aquarium                                   | \$308                       |
| Shedd historic yolk samples (bomb calorimetry)                                  | 34 yolk samples x \$44/sample                                   | \$1,496               | \$                                          |                                                                        |                                                     | \$1,496                     |
| Shell composites (bomb calorimetry)                                             | 5 composites x \$44/sample                                      | \$220                 | \$                                          |                                                                        |                                                     | \$220                       |
| Internal component composites (bomb calorimetry)                                | 5 composites x \$44/sample                                      | \$220                 | \$                                          |                                                                        |                                                     | \$220                       |
| Food items without historical calorimetry data                                  | 6 items x 3 replicates x \$44/sample                            | \$792                 | \$                                          |                                                                        |                                                     | \$792                       |
| AOP diet switch prey items                                                      | Diet items: \$4.60/lb x 277.2 lb food for 3 sharks for 3 months | \$1,276               | \$                                          |                                                                        |                                                     | \$1,276                     |
| AOP yolks collected during diet study (bomb calorimetry + nutritional analysis) | 73 yolks/6 months of monitoring x \$98/sample                   | \$7,154               | \$                                          |                                                                        |                                                     | \$7,154                     |
| Shipping                                                                        | \$50/shipment x 17 shipments                                    | \$850                 | \$                                          |                                                                        |                                                     | \$850                       |



## **BUDGET JUSTIFICATION**

State why each item listed in the Budget Items Form costs what it does and indicate how each item relates to the proposed scope of work. Sufficient detail should be provided to address any potential reviewer concerns with respect to cost and need. If the lifespan of equipment costing over \$500 exceeds that of the grant period, detail and justify who will assume ownership of the item. **LIMIT 2,500 CHARACTERS with spaces**

### Egg energetics

- GAI has previously use NPAL labs for bomb calorimetry analysis with a price per sample \$44.
- Yolk samples will be analyzed from 50% of the total yolked eggs laid at each institution based on records from previously sampled eggs. Number of yolk samples analyzed varied by institution as fecundity was not consistent across aquariums.
- Composite samples of shells from 5 different females will be analyzed to obtain a mean caloric value for shells
- Composite samples of internal material from 5 different females will be analyzed to obtain a mean caloric value for internal material
- Shipping, including dry ice (\$50 per box for analysis x 4 shipments = \$200)

### Diet energetics

- Food items
  - o We estimate that there could be 4-6 food items that are sourced from non-traditional vendors.
  - o Additional food items to be analyzed are: 6 items x 3 replicates/item x \$44/sample = \$792
- Shipping. Diet items collected by institutions will be shipped to GAI or Ripley's for collation (\$50 per box x 4 shipments= \$200). Diet items will be sent to the laboratory in batch (\$50 per box for analysis x 1 shipment = \$100).

### Diet switch study

- Specific invertebrate items may need to be purchased to cover the increase in invertebrates in the diet.
- o Sharks are fed ~7.7lb/week x 3 sharks x 12 weeks x \$4.60/lb food = \$1,276
- In addition to total gross energy, yolks collected during the diet switch experiment will undergo further analysis to include dry matter, crude protein, fat, and ash for a deeper examination of egg nutrient quality with change in prey items. Sample cost to include these additional parameters is \$98/sample.
  - o Estimated eggs to be collected over 6 months is 73 samples x \$98 = \$7,154
- Shipping: Egg shipments will occur monthly for six months (\$50/box x 6 shipments = \$300). Yolk samples will be sent to the lab in two batches (\$50/box x 2 shipments = \$100)

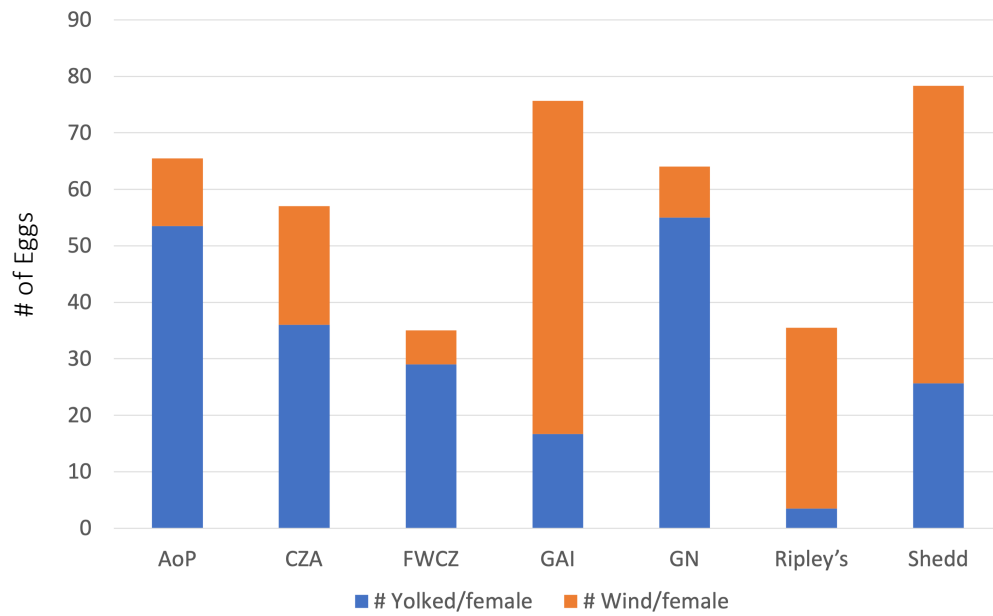
**Appendix F. Additional Collaborators**

Golden Nugget, Statement of Institutional Support form attached.

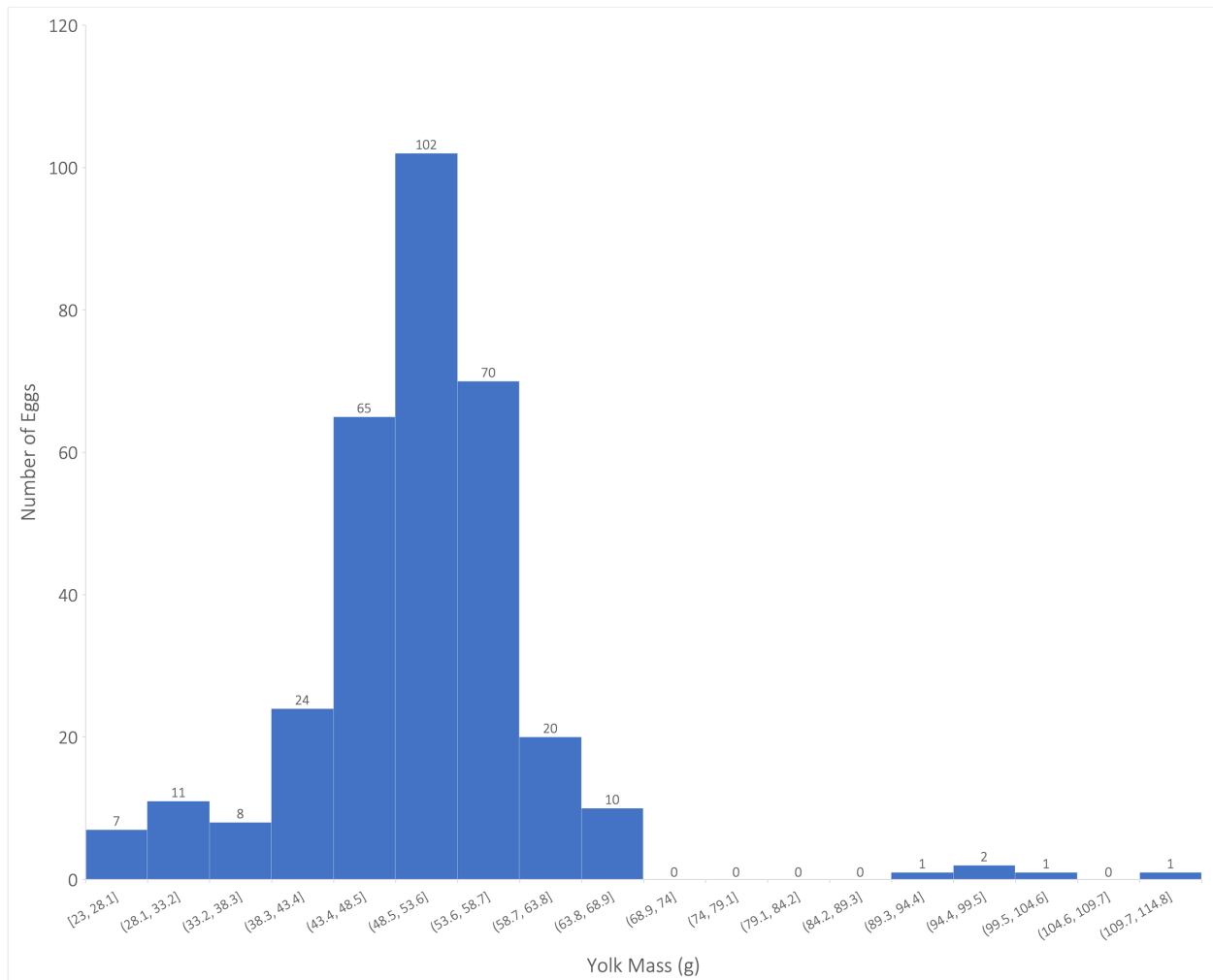
Columbus Zoo and Aquarium, Statement of Institutional Support form attached.

**Appendix A.** List of institutions that contributed eggs to the reproductive monitoring study for one year's time. Half of the yolk samples collected from these animals will be analyzed as part of goal 1.

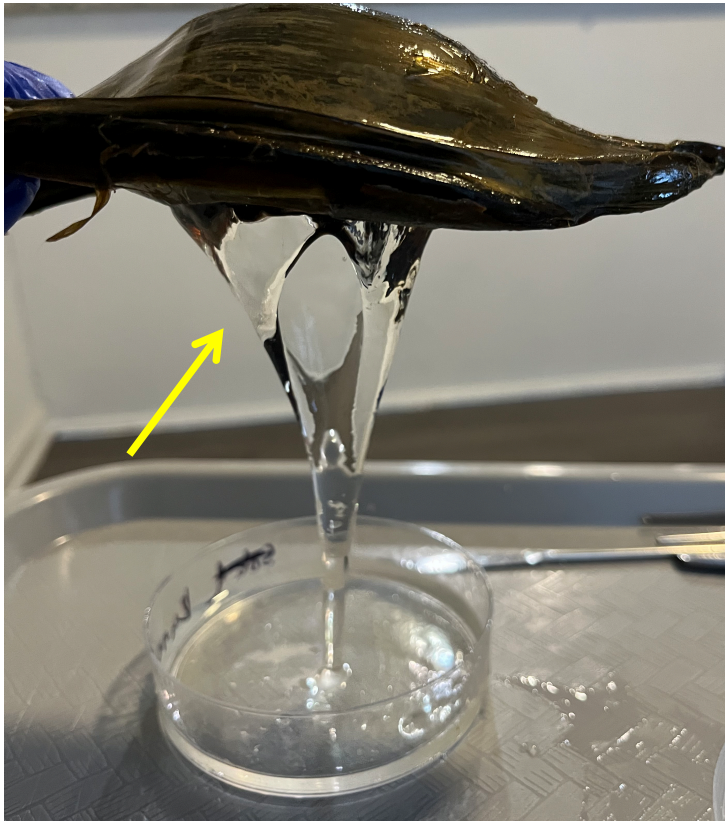
| Institution                 | Number of Females | Number of yolked eggs contributed | Number of wind eggs contributed | Number of total eggs |
|-----------------------------|-------------------|-----------------------------------|---------------------------------|----------------------|
| Aquarium of the Pacific     | 4                 | 214                               | 48                              | 262                  |
| Columbus Zoo and Aquarium   | 2                 | 72                                | 42                              | 114                  |
| Fort Wayne Children's Zoo   | 1                 | 29                                | 6                               | 35                   |
| Georgia Aquarium            | 3                 | 50                                | 177                             | 227                  |
| Golden Nugget Las Vegas     | 1                 | 55                                | 9                               | 64                   |
| Ripley's Aquarium of Canada | 2                 | 7                                 | 64                              | 71                   |
| Shedd Aquarium              | 3                 | 77                                | 158                             | 235                  |



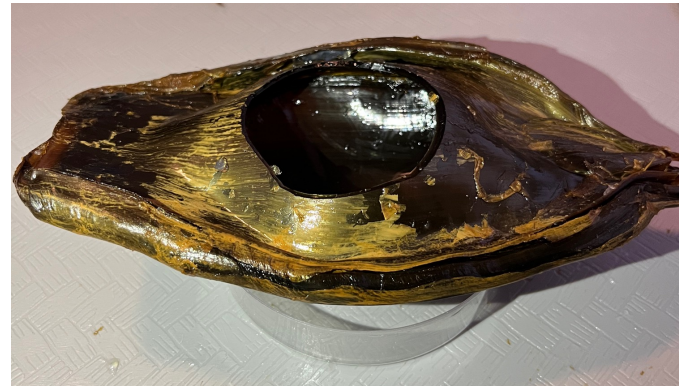
**Appendix B.** Graphical display of number of eggs per female by type (yolk: blue or wind: orange) over the course of one complete year of monitoring.



**Appendix C.** Histogram of distribution of yolk mass from eggs monitored for one year's time. Numbers above the bars indicated the sample size for each mass range



Internal components indicated by yellow arrow in wind eggs



Shell with internal components removed



Example of egg case with yolk present

**Appendix D.** Three components of an egg that will be analyzed by bomb calorimetry to obtain total cost of each egg: internal components (left side), shell (top right) and yolk (bottom right).

## Appendix E. Milestones and timelines to complete each aspect of the project.

[illegible]

## Citations

Mull, C. G., Lowe, C. G., & Young, K. A. (2008). Photoperiod and water temperature regulation of seasonal reproduction in male round stingrays (*Urobatis halleri*). *Comparative Biochemistry and Physiology Part A: Molecular & Integrative Physiology*, 151(4), 717-725.

Traylor-Holzer, K. 2021. *Population Viability Analysis (PVA) Report for Population Augmentation of Zebra Sharks (Stegostoma tigrinum)* in Raja Ampat, Indonesia. Apple Valley, MN: IUCN SSC Conservation Planning Specialist Group.

Wade, G. N., Schneider, J. E., & Li, H. Y. (1996). Control of fertility by metabolic cues. *American Journal of Physiology-Endocrinology And Metabolism*, 270(1), E1-E19.

Watson, L., & Janse, M. (2017). Reproduction and husbandry of zebra sharks, *Stegostoma fasciatum*, in aquaria. *The Elasmobranch Husbandry Manual II: Recent Advances in the Care of Sharks, Rays and their Relatives*. Columbus, Ohio: Ohio Biological Survey Inc, 421-432.

**Lance Adams**  
CURRICULUM VITAE  
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**Current Employment**

2001 Senior Veterinarian, Aquarium of the Pacific

**Education**

2001 Aquatic Animal Medicine Internship, New England Aquarium  
2000 Rotating Small Animal Medicine and Surgery Internship, VCA West Los Angeles Animal Hospital  
1999 DVM, Kansas State University  
1995 B.S. in Animal Science, California State Polytechnic University

**Experience**

1998 Externship in Zoological Medicine, Sea World San Diego  
1998 Rotation in Zoological Medicine, KSU Exotic Animal, Wildlife, and Zoo Animal Medicine Service  
1998 Externship in Zoological Medicine, Denver Zoo  
1997 Externship in Wildlife Medicine, HOWL Wildlife Center

**Professional Memberships**

American Veterinary Medical Association, member  
International Association of Aquatic Animal Medicine, member  
American Association of Zoo Veterinarians, member  
Association of Zoos & Aquariums, member

**Select Publications**

- Adams, LA**, K Lyons, J Monday, E Larkin, and JT Wyffels. (2023). Cost of parthenogenesis on growth and longevity in ex situ zebra sharks *Stegostoma tigrinum*. *Endangered Species Research* 50:81-91. DOI: 10.3354/esr01224
- Adams, LA**, K Lyons, E Larkin, N Leier, J Monday, C Plante, JM Dubach, and JT Wyffels. (2022). Artificial insemination and parthenogenesis in the Zebra Shark *Stegostoma tigrinum*. *Frontiers in Marine Science* 9, 886616. DOI: 10.3389/fmars.2022.886616
- Wyffels JT, **L Adams**, F Bulman, A Fustukjian, MW Hyatt, KA Feldheim, and LM Penfold. (2021). Artificial insemination and parthenogenesis in the whitespotted bamboo shark *Chiloscyllium plagiosum*. *Scientific Reports* 11, 9966. DOI: 10.1038/s41598-021-88568-y
- Wyffels JT, R George, **L Adams**, C Adams, T Clauss, A Newton, MW Hyatt, C Yach, and LM Penfold. (2020). Testosterone and semen seasonality for the sand tiger *Carcharias taurus*. *Biology of Reproduction* 102: 876-887. DOI: 10.1093/biolre/iox221
- Peel, MJ, **L Adams**, B Stevens, MM Garner, C Maguire (2021). Systemic phaeohyphomycosis in a splitose rockfish (*Sebastes diploproa*) caused by *Devriesia* sp. *Journal of Fish Diseases* 44(5): 639-644.
- Gunther-Harrington, CT, ME Moriarty, CL Field, **L Adams**, CK Johnson, MJ Murray. Transthoracic echocardiographic evaluation and serum cardiac troponin values in anesthetized healthy female southern sea otters (*Enhydra lutris nereis*). *Journal of Zoo and Wildlife Medicine*, 52(2): 490-498.
- Moriarty, ME, MA Miller, MJ Murray, P Duignan, CT Gunther-Harrington, CL Field, **L Adams**, TL Schmitt, CK Johnson (2021). Exploration of serum cardiac troponin I as a biomarker of cardiomyopathy in southern sea otters. Exploration of serum cardiac troponin I as a biomarker of cardiomyopathy in southern sea otters. *American Journal of Veterinary Research*. 82(7): 529-537.
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- Schmidt, TL, S Munn, **L Adams**, J Hicks (2013). The use of spirometry to evaluate pulmonary function in olive ridley sea turtles (*Lepidochelys olivacea*) with positive buoyancy disorders. *Journal of Zoo and Wildlife Medicine*. 50(3):368-376.
- Bonar, CJ, MM Garner, ES Weber III, CJ Keller, M Murray, **L Adams**, S Frasca Jr. (2013). Pathologic Findings in Weedy (*Phyllopteryx taeniolatus*) and Leafy (*Phycodurus eques*) Seadragons. *Veterinary Pathology*. 50(3): 368-376.
- LaDouceur, EEB, MM Garner, J Wynne, S Fish, **L Adams** (2013). Ulcerative Umbrellar Lesions in Captive Moon Jelly (*Aurelia aurita*) Medusae. *Veterinary Pathology*. 50(3): 434-442.
- Garner, MM, SD Atkinson, SL Hallett, JL Bartholomew, RW Nordhausen, H Reed, **L Adams**, B Whitaker (2007). Renal myxozoanosis in weedy sea dragons, *Phyllopteryx taeniolatus* (Lacepede), cause by *Sinuolinea phyllopteryxa* n. sp. *Journal of Fish Diseases*. 29: 1-9.
- Sherrill, J, GM Peavy, MJ Kopit, MM Garner, CH Gardiner, **L Adams** (2004). Use of Laser Rhinoscopy to Treat a Nasal Obstruction in a Captive California Sea Lion (*Zalophus californianus*). *Journal of Zoo and Wildlife Medicine*. 35(2): 232-241.
- Adams, L**, R Isaza, E Greiner (2001). Fatal Pentastomiasis in Captive African Dwarf Crocodile Hatchlings (*Osteolaemus tetraspis*). *Journal of Zoo and Wildlife Medicine* 32(4), 500-502.

**Kady Lyons**  
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**Current Employment**

2019-present      Research Scientist, Georgia Aquarium

**Education**

2013-2018      Ph.D. in Physiology, University of Calgary  
2009-2013      M.S. in Biology, California State University Long Beach  
2005-2009      B.S. in Marine Biology, University of California Santa Cruz

**Experience**

2018-2019      Postdoctoral researcher, CSULB  
2010-2013      CSULB and Monterey Bay Aquarium white shark monitoring project coordinator  
2018 Spring      *Lecturer:* Human Biology Course Instructor, CSULB  
2014 – 2017      *Writing Tutor:* Writing Support Center, U of Calgary  
2013 Fall      *Teaching Assistant:* Introduction to Developmental Biology, U of Calgary  
2012 – 2013      *Biologist* aboard 3-day ocean liveaboard high school AP Environmental Studies semester trips  
2009 – 2013      *Teaching Assistant:* Human Biology Lab Instructor, CSULB

**Selected Grants & Awards**

2021 AZA Conservation Grant Fund recipient  
2020 Save Our Seas Foundation Small Grant Recipient  
2018 South Carolina State Wildlife Grants Project (co-author)  
2016 Eugenie Clark award for young professional female scientist (American Elasmobranch Society)  
2016 P.E.O. scholarship  
2014 Save Our Seas Foundation Small Grant Fund recipient  
2014 Outstanding Master's thesis award for College of Natural Science and Mathematics (CSULB)  
2014 Vanier Scholarship recipient  
2013 Alberta Innovates-Technology Futures Doctoral Scholarship  
2013 Gruber Award for best student oral presentation (American Elasmobranch Society)

**Selected Publications**

Freedman, RM, Anderson, JM, Caldow, C, Stirling, B, Rex, P, Spurgeon, E, McCullough, S, **Lyons, K**, May, J, White, CF, Logan, RK, Meese, E, Burns, ES, Clevensine, AJ, O'Sullivan, J, Winkler, C, Duncan, L, Cajandig, M, Lowe, CG. Evidence of Juvenile White Sharks (*Carcharodon carcharias*) habitat use at the north Channel Islands. Submitted. *Journal of Fish Biology*.

Adams, LA, **Lyons, K**, Monday, J, Larkin, E, Wyffels, JT. Costs of parthenogenesis on growth and longevity in ex situ Zebra Sharks (*Stegostoma tigrinum*). 2023. *Endangered Species Research*. 50: 81-91.  
Doi.org/10.3354/esr01224

Adams, LA, **Lyons, K**, Larkin, E, Leier, N, Monday, J, Plante, C, Dubach, JM, Wyffels, JT. Artificial insemination and parthenogenesis in the Zebra Shark *Stegostoma tigrinum*. 2022. *Frontiers in Marine Science*. 9:886616.  
Doi:10.3389/fmars.2022.886616

O'Sullivan, J, Lowe, GC, Sosa-Nishizaki, S, Jorgensen, S, Anderson, J, Farrugia, Garcia-Rodriguez, E, **Lyons, K**, McKinzie, M, Onate-Gonzalez, E, Weng, K, White, C, Winkler, C., Van Houtan, K. 2022. A biologging database of juvenile white sharks from the Northeast Pacific. Scientific Data. #SDATA-21-01167B.

Anderson, J, Clevensine, AJ, Stirling, BS, Burns, ES, Meese, EN, White, CF, Logan, RK, O'Sullivan, J, Rex, PT, May, J, **Lyons, K**, Winkler, C, García-Rodríguez, E, Sosa-Nishizaki, O, Lowe, CG. 2021. Non-random Co-occurrence of Juvenile White Sharks (*Carcharodon carcharias*) at Seasonal Aggregation Sites in Southern California. *Frontiers in Marine Science*:1095.

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- van Zinnicq Bergmann, MPM, Postaire, BD, Gastrich, K, Heithaus, MR, Hoopes, LA, **Lyons, K**, Papastamatiou, YP, Schneider, EVC, Strickland, BA, Talwar, BS, Chapman, DD, Bakker, J. Elucidating shark diets with DNA metabarcoding from cloacal swabs. 2021. *Molecular Ecology Resources*, 21(4): 1056-1067. doi: 10.1111/1755-0998.13315.
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- White, CF, **Lyons, K**, Jorgensen, SJ, O'Sullivan, JB, Winkler, C, Weng, KC, Lowe, CG. 2019. Quantifying habitat selection and variability in habitat suitability for juvenile white sharks. PLoS ONE 14(5): e0214642. <https://doi.org/10.1371/journal.pone.0214642>
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- Lyons, K**, Wynne-Edwards, KE. 2019. Steroid concentrations in maternal serum and uterine histotroph in round stingrays (*Urobatis halleri*). General and Comparative Endocrinology. 274:8-16. doi.org/10.1016/j.ygcen.2018.12.009
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- Lyons, K**, Burket, SR, Brooks, BW. 2018. Sex may influence environmental diphenhydramine accumulation in Round Stingrays. *Marine Pollution Bulletin* 135: 648-653. <https://doi.org/10.1016/j.marpolbul.2018.07.042>

## Current Employment

2023 Research Scientist, Ripley's Aquariums

## Education

2013-2014 Certificate of Bioinformatics and Computational Biology, University of Delaware  
2004-2005 Post-doctorate in Developmental Biology, Aichi Medical University  
1991-2001 Ph.D. in Physiology, Clemson University  
2005-2009 B.S. in Biology with minors in Chemistry & Religion, Bradley University

## Experience

2023-present Research Scientist, Ripley's Aquariums  
2016-present Scholar, University of Delaware, Center for Bioinformatics and Computation Biology  
2015-2020 Research Scientist, South-East Zoo Alliance for Reproduction & Conservation  
2012-2014 National Institutes of Health Fellowship, University of Delaware  
2007-2015 Research Scientist, Daemen College Center for Wound Healing Research  
2006-2007 Biology Instructor, Daemen College  
2004-2005 Postdoctoral researcher, Aichi Medical University, Japan

## Selected Grants & Awards

2020 Ichthyology Society of Japan  
2019 Distinguished Service Award, American Elasmobranch Society  
2019 Henry Doorly Zoo and Aquarium  
2018 Morris Animal Foundation Grant  
2018 SeaWorld & Busch Gardens Conservation Fund Grant  
2017 SeaWorld & Busch Gardens Conservation Fund Grant  
2012 NIH Fellowship  
2011 Department of Defense CDMRP/DMRDP Grant  
2010 Department of Defense USAMRMC Grant  
2004 National Science Foundation JSPS Fellowship

## Service

American Elasmobranch Society  
Board of Directions (2020 – 2024)  
Outreach & Education committee (2019-2023)  
Equity & Diversity Committee (2016-2018, Chair 2019)  
Executive Committee (2013-2017)  
Nominating committee (1997, 2000, 2006 & Chair 2018)  
Student Activities Council (Chair 1999-2000, 2001)  
*Stegostoma tigrinum* Augmentation and Restoration (StAR) Project  
Husbandry Committee (2020-present)  
Reproduction and Artificial Insemination ad hoc Committee (2020-present)  
Neotropical Ichthyology, Associate Editor (2017-present)  
Little Skate Genome Project, Key Personnel (2012-present)

## Select Publications

Adams, LA, K Lyons, J Monday, E Larkin, and **JT Wyffels**. (2023). Cost of parthenogenesis on growth and longevity in ex situ zebra sharks *Stegostoma tigrinum*. *Endangered Species Research* 50:81-91. DOI: 10.3354/esr01224  
Feldheim, KA, **JT Wyffels** and K Lyons. (2022). The role of aquaria in the advancement of elasmobranch reproductive biology. *Frontiers in Marine Science* 9, 963542. DOI: 10.3389/fmars.2022.963542  
**Wyffels, JT**, R George, EF Christiansen, TM Clauss, AL Newton, MW Hyatt, C Buckner, V LePage, FE Latson, and LM Penfold. (2022). Reproductive cycle and periodicity of in situ and aquarium female sand tiger sharks

*Carcharias taurus* from the western North Atlantic. *Frontiers in Marine Science* 9, 925749. DOI: 10.3389/fmars.2022.925749

- Adams, LA, K Lyons, E Larkin, N Leier, J Monday, C Plante, JM Dubach, and **JT Wyffels**. (2022). Artificial insemination and parthenogenesis in the Zebra Shark *Stegostoma tigrinum*. *Frontiers in Marine Science* 9, 886616. DOI: 10.3389/fmars.2022.886616
- Wyffels JT**, L Adams, F Bulman, A Fustukjian, MW Hyatt, KA Feldheim, and LM Penfold. (2021). Artificial insemination and parthenogenesis in the whitespotted bamboo shark *Chiloscyllium plagiosum*. *Scientific Reports* 11, 9966. DOI: 10.1038/s41598-021-88568-y
- Yamaguchi, A, K Furumitsu, & **J Wyffels**. (2021). Reproductive Biology and Embryonic Diapause as a Survival Strategy for the East Asian Endemic Eagle Ray *Aetobatus narutobiei*. *Frontiers in Marine Science* 8, 768701. DOI: 10.3389/fmars.2021.768701
- Wyffels JT**, R George, L Adams, C Adams, T Clauss, A Newton, MW Hyatt, C Yach, and LM Penfold. (2020). Testosterone and semen seasonality for the sand tiger *Carcharias taurus*. *Biology of Reproduction* 102: 876-887. DOI: 10.1093/biolre/ioz221
- Wyffels JT**, C Coco, C Schreiber, D Palmer, T Clauss, F Bulman, R George, C Pelton, K Feldheim, and T Handsel. (2020). Natural environmental conditions and collaborative efforts provide the secret to success for sand tiger shark *Carcharias taurus* reproduction in aquaria. *Zoo Biology* 39: 355-363. DOI: 10.1002/zoo.21558
- Mylniczenko ND, S Sumigama S, **JT Wyffels**, CJ Wheaton, TL Guttridge, S DiRocco, and LM Penfold. (2020). Ultrasonographic and hormonal characterization of reproductive health and disease in wild, semiwild, and aquarium-housed southern stingrays (*Hypanus americanus*). *American Journal of Veterinary Research* 80:931-942. DOI: 10.2460/ajvr.80.10.931
- Furumitsu K, **JT Wyffels**, and A Yamaguchi. (2019) Reproduction and embryonic development of the red stingray *Hemirhamphysa akajei* from Ariake Bay, Japan. *Ichthyology Research* 66:419-436. DOI: 10.1007/s10228-019-00687-9
- Wyffels JT**, BL King, J Vincent, C Chen, CH Wu, and SW Polson. (2014) SkateBase, an elasmobranch genome project and collection of molecular resources for chondrichthyan fishes. *F1000Research* 3:191. DOI: 10.12688/f1000research.4996.1
- Wyffels JT**, AB Bodine, CA Luer, and CJ Walsh. (2007). In vivo exposure of clearnose skates, *Raja eglanteria*, to ionizing X-radiation: Acute effects on the peripheral blood, spleen, and epigonal and Leydig organs. *Fish and Shellfish Immunology* 23:401-418. DOI: 10.1016/j.fsi.2006.12.002
- Luer, CA, CJ Walsh, **JT Wyffels**, and AB Bodine. (2007). Normal embryonic development in the clearnose skate, *Raja eglanteria*, with experimental observations on artificial insemination. *Environmental Biology of Fishes* 80:239-255. DOI: 10.1007/s10641-007-9219-4
- Wyffels JT**, M Masuda, J Sakai, and Y Itoh. (2006). Characteristics of *Scyliorhinus torazame* egg case jelly. *Journal of the Japanese Society for Elasmobranch Studies* 45:1-7.

## Book Chapters

- Wyffels JT** and LM Penfold. (2022). Molecular Aspects of Elasmobranch Reproduction and Artificial Insemination. In: *Biology of Sharks & Their Relatives*, Edition 3. J Carrier, C Simpfendorfer, M Heithaus and K Yopak (Eds), CRC Press. In Press. pp. 231-250. DOI: 10.1201/9781003262190
- Luer CA and **JT Wyffels**. (2022). Developmental Biology of Elasmobranch Fishes: Updates and Selected Topics. In: *Biology of Sharks & Their Relatives*, Edition 3. J Carrier, C Simpfendorfer, M Heithaus and K Yopak (Eds), CRC Press. In Press. pp. 251-288. DOI: 10.1201/9781003262190
- Penfold LM, **JT Wyffels**. (2019). Reproductive Science in Sharks and Rays. In: *Reproductive Sciences in Animal Conservation. Advances in Experimental Medicine and Biology*, vol 1200, P Comizzoli. J Brown and W Holt (Eds.), Springer, Cham., pp. 465-488. DOI: 10.1007/978-3-030-23633-5\_15
- Wyffels JT**. (2009). Embryonic development of chondrichthyan fishes – A. review. In: *Development of Non-Teleost Fishes*. Y. Kunz, BG Kapoor and CA Luer (Eds.), Science Publishers Inc., Enfield, pp. 1-103. DOI: 10.1201/b10184-2

## 2023 AZA CONSERVATION GRANTS FUND

### Statement of Institutional Support

One copy of this statement must be **electronically signed** by the institutional CEO/Director using the checkbox below, and submitted as an attachment to the CGF Application email titled **“SIS Institution Name”** for *each* institution or organization that fits one or more of the following criteria:

1. **Has an employee listed as a Principal Investigator or Co-Investigator on the Application Cover Sheet.**
2. **Is listed as a collaborating institution or organization on the Application Cover Sheet.**

By signing this form, all participating institutions, whether participating as a facility or through an employee, take responsibility for: 1) completing their part of the proposed project and 2) obtaining Institutional Animal Care and Use Committee (IACUC)\* or Institutional Review Board (IRB)\*\* approvals, if necessary. All accountability for conducting responsible research lies in the hands of each participating institution, not the CGF Scientific Advisory Committee.

All *Statement of Institutional Support* forms must be received as part of the CGF Application Email.

The (*Institution Name*) Aquarium of the Pacific has reviewed the proposal titled (*Proposal Title*) Maximizing fecundity through energy optimization: Is nutrition limiting egg laying habits of zebra sharks? and commits itself to providing the time, resources and/or financial support as stated in the application, and for obtaining IACUC and/or IRB approvals, if necessary, should the proposal be selected for funding by AZA's Conservation Grants Fund.

Dr. Peter Kareiva

Name

CEO and President, Aquarium of Pacific

Title

(Zoos and Aquariums must include the signature of the Institutional CEO/Director above and cannot substitute the signature of the COO, General Curator, Vice President, Department Director, or Principal Investigator. Universities, Colleges, and Federal or State Government entities may substitute the signature of a Governing Official.)

2/20/2023

Date

☒ **Checking this box serves as my electronic signature to this letter of support. I affirm that I am the person listed above and understand that this letter WILL NOT be accepted unless the box is checked.**

\* Institutional Animal Care and Use Committee (IACUC, for U.S. applicants) or the equivalent oversight committee (for non-U.S. applicants) approval is desirable for any proposal, but is required for any project in which animals are to be involved in invasive procedures and/or procedures that may result in pain, harm, distress, and/or altering of behavior (this includes any time an animal in the wild is sedated or anesthetized). This generally excludes samples obtained during regularly scheduled veterinary or husbandry procedures or emergency medical procedures. IACUC (or the international equivalent thereof) approval is required for both *in situ* and *ex situ* projects. Each institution involved in a project must submit the project proposal to its IACUC (or equivalent) for review. In order to be considered complete, proposals must include the Statement of Institutional Support from each participating institution/organization as recognition of compliance with these IACUC (or equivalent) approval guidelines.

If an institution is not registered as a research facility or similar facility and therefore does not have an IACUC (or equivalent), it needs to become affiliated with a university, zoo, or other body that has an IACUC (or equivalent) in order to obtain approval.

\*\* Any projects supported through CGF funding that involve human subjects must be conducted under the auspices of a federally registered Institutional Review Board (IRB) and be consistent with 45 CFR Part 690: Federal Policy for the Protection of Human Subjects if the work or researchers are based in the USA. For members submitting from outside the USA for projects to be completed entirely outside the USA and its territories may use a process that meets the requirements of the prevailing legislation for protection of human subjects where the principal investigator is based or in the country where data collection will occur, whichever is more stringent. We note that all protocols must still conform to the types of protections outlined in 45 CFR Part 690 irrespective of the reviewing organization.

# CONSERVATION GRANTS FUND

Association of Zoos & Aquariums

## 2023 AZA CONSERVATION GRANTS FUND

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All *Statement of Institutional Support* forms must be received as part of the CGF Application Email.

The (*Institution Name*) Columbus Zoo and Aquarium has reviewed the proposal titled (*Proposal Title*) Maximizing fecundity through energy optimization: Is nutrition limiting egg laying habits of zebra sharks? and commits itself to providing the time, resources and/or financial support as stated in the application, and for obtaining IACUC and/or IRB approvals, if necessary, should the proposal be selected for funding by AZA's Conservation Grants Fund.

\_\_\_\_\_  
Name

Tom Schmid



\_\_\_\_\_  
Title

CEO

(Zoos and Aquariums must include the signature of the Institutional CEO/Director above and cannot substitute the signature of the COO, General Curator, Vice President, Department Director, or Principal Investigator. Universities, Colleges, and Federal or State Government entities may substitute the signature of a Governing Official.)

\_\_\_\_\_  
Date

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## 2023 AZA CONSERVATION GRANTS FUND

### Statement of Institutional Support

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All *Statement of Institutional Support* forms must be received as part of the CGF Application Email.

The (*Institution Name*) Fort Wayne Children's Zoo has reviewed the proposal titled (*Proposal Title*) Maximizing fecundity through energy optimization: Is nutrition limiting egg laying habits of zebra sharks? and commits itself to providing the time, resources and/or financial support as stated in the application, and for obtaining IACUC and/or IRB approvals, if necessary, should the proposal be selected for funding by AZA's Conservation Grants Fund.

Rick Schuiteman

Name

Executive Director

Title

(Zoos and Aquariums must include the signature of the Institutional CEO/Director above and cannot substitute the signature of the COO, General Curator, Vice President, Department Director, or Principal Investigator. Universities, Colleges, and Federal or State Government entities may substitute the signature of a Governing Official.)

February 3, 2023

Date

☒ **Checking this box serves as my electronic signature to this letter of support. I affirm that I am the person listed above and understand that this letter WILL NOT be accepted unless the box is checked.**

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## 2023 AZA CONSERVATION GRANTS FUND

### Statement of Institutional Support

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All *Statement of Institutional Support* forms must be received as part of the CGF Application Email.

The (*Institution Name*) Georgia Aquarium has reviewed the proposal titled (*Proposal Title*) Maximizing fecundity through energy optimization: Is nutrition limiting egg laying habits of zebra sharks? and commits itself to providing the time, resources and/or financial support as stated in the application, and for obtaining IACUC and/or IRB approvals, if necessary, should the proposal be selected for funding by AZA's Conservation Grants Fund.

Dr. Brian Davis

Name

President and Chief Executive Officer

Title

(Zoos and Aquariums must include the signature of the Institutional CEO/Director above and cannot substitute the signature of the COO, General Curator, Vice President, Department Director, or Principal Investigator. Universities, Colleges, and Federal or State Government entities may substitute the signature of a Governing Official.)

2-8-2023

Date

☒ **Checking this box serves as my electronic signature to this letter of support. I affirm that I am the person listed above and understand that this letter WILL NOT be accepted unless the box is checked.**

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Association of Zoos & Aquariums

## 2023 AZA CONSERVATION GRANTS FUND

### Statement of Institutional Support

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All *Statement of Institutional Support* forms must be received as part of the CGF Application Email.

The (*Institution Name*) Golden Nugget Las Vegas has reviewed the proposal titled (*Proposal Title*) Maximizing fecundity through energy optimization: Is nutrition limiting egg laying habits of zebra sharks? and commits itself to providing the time, resources and/or financial support as stated in the application, and for obtaining IACUC and/or IRB approvals, if necessary, should the proposal be selected for funding by AZA's Conservation Grants Fund.

JAMES PRAPPAS

Name

Director of Animal Operations

Title

(Zoos and Aquariums must include the signature of the Institutional CEO/Director above and cannot substitute the signature of the COO, General Curator, Vice President, Department Director, or Principal Investigator. Universities, Colleges, and Federal or State Government entities may substitute the signature of a Governing Official.)

February 5, 2023

Date

☒ **Checking this box serves as my electronic signature to this letter of support. I affirm that I am the person listed above and understand that this letter WILL NOT be accepted unless the box is checked.**

\* Institutional Animal Care and Use Committee (IACUC, for U.S. applicants) or the equivalent oversight committee (for non-U.S. applicants) approval is desirable for any proposal, but is required for any project in which animals are to be involved in invasive procedures and/or procedures that may result in pain, harm, distress, and/or altering of behavior (this includes any time an animal in the wild is sedated or anesthetized). This generally excludes samples obtained during regularly scheduled veterinary or husbandry procedures or emergency medical procedures. IACUC (or the international equivalent thereof) approval is required for both *in situ* and *ex situ* projects. Each institution involved in a project must submit the project proposal to its IACUC (or equivalent) for review. In order to be considered complete, proposals must include the Statement of Institutional Support from each participating institution/organization as recognition of compliance with these IACUC (or equivalent) approval guidelines.

If an institution is not registered as a research facility or similar facility and therefore does not have an IACUC (or equivalent), it needs to become affiliated with a university, zoo, or other body that has an IACUC (or equivalent) in order to obtain approval.

\*\* Any projects supported through CGF funding that involve human subjects must be conducted under the auspices of a federally registered Institutional Review Board (IRB) and be consistent with 45 CFR Part 690: Federal Policy for the Protection of Human Subjects if the work or researchers are based in the USA. For members submitting from outside the USA for projects to be completed entirely outside the USA and its territories may use a process that meets the requirements of the prevailing legislation for protection of human subjects where the principal investigator is based or in the country where data collection will occur, whichever is more stringent. We note that all protocols must still conform to the types of protections outlined in 45 CFR Part 690 irrespective of the reviewing organization.

## 2023 AZA CONSERVATION GRANTS FUND

### Statement of Institutional Support

One copy of this statement must be **electronically signed** by the institutional CEO/Director using the checkbox below, and submitted as an attachment to the CGF Application email titled "**SIS Institution Name**" for *each* institution or organization that fits one or more of the following criteria:

1. **Has an employee listed as a Principal Investigator or Co-Investigator on the Application Cover Sheet.**
2. **Is listed as a collaborating institution or organization on the Application Cover Sheet.**

By signing this form, all participating institutions, whether participating as a facility or through an employee, take responsibility for: 1) completing their part of the proposed project and 2) obtaining Institutional Animal Care and Use Committee (IACUC)\* or Institutional Review Board (IRB)\*\* approvals, if necessary. All accountability for conducting responsible research lies in the hands of each participating institution, not the CGF Scientific Advisory Committee.

All *Statement of Institutional Support* forms must be received as part of the CGF Application Email.

The (*Institution Name*) Ripley's Aquarium of Canada has reviewed the proposal titled (*Proposal Title*) Maximizing fecundity through energy optimization: Is nutrition limiting egg laying habits of zebra sharks? and commits itself to providing the time, resources and/or financial support as stated in the application, and for obtaining IACUC and/or IRB approvals, if necessary, should the proposal be selected for funding by AZA's Conservation Grants Fund.

Name

Jim Handzel

Title

V.P. Husbandry / Ripley Entertainment

(Zoos and Aquariums must include the signature of the Institutional CEO/Director above and cannot substitute the signature of the COO, General Curator, Vice President, Department Director, or Principal Investigator. Universities, Colleges, and Federal or State Government entities may substitute the signature of a Governing Official.)

Date

2/2/2023

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## 2023 AZA CONSERVATION GRANTS FUND

### Statement of Institutional Support

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1. **Has an employee listed as a Principal Investigator or Co-Investigator on the Application Cover Sheet.**
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All *Statement of Institutional Support* forms must be received as part of the CGF Application Email.

The (*Institution Name*) Shedd Aquarium has reviewed the proposal titled (*Proposal Title*) Maximizing fecundity through energy optimization: Is nutrition limiting egg laying habits of zebra sharks? and commits itself to providing the time, resources and/or financial support as stated in the application, and for obtaining IACUC and/or IRB approvals, if necessary, should the proposal be selected for funding by AZA's Conservation Grants Fund.

Dr. Bridget Coughlin, PhD

Name

President and CEO

Title

(Zoos and Aquariums must include the signature of the Institutional CEO/Director above and cannot substitute the signature of the COO, General Curator, Vice President, Department Director, or Principal Investigator. Universities, Colleges, and Federal or State Government entities may substitute the signature of a Governing Official.)

2/7/2023

Date

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## 2023 AZA CONSERVATION GRANTS FUND

### Statement of SAFE Species Program Alignment

AZA's Conservation Grants Fund awards funds to members to implement SAFE program plans and advance the species program team's priorities. Because of the granting program's alignment with SAFE species programs, coordination with SAFE Species Programs is expected.

Program leaders are asked to submit this form for proposals related to their SAFE species program. To assure program review of relevant submissions, AZA SAFE Program Leaders may also be asked to serve as expert reviewers during the application review process and may express their support for a proposal at that time.

One copy of this statement must be **electronically signed** by the SAFE Species Program Leader using the checkbox below, and submitted as an attachment to the Application email titled **"SOA SAFE Species Program Name"** for each SAFE species program relevant to the Application.

All *Statement of SAFE Species Program Alignment* forms must be received as part of the CGF Application Email.

I affirm that, to the best of my knowledge, the proposal titled (*Proposal Title*) Maximizing fecundity through energy optimization: Is nutrition limiting egg laying habits of zebra sharks? aligns with and advances the objectives and priorities of SAFE (*Species Program Name*) Sharks and Rays.

Hap Fatzinger  
Name

NC Aquarium Director/SAFE Sharks and Rays Program Leader  
Title

NC Aquarium  
Organization

Sharks and Rays  
SAFE Species Program

3/7/2023  
Date

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